

Paper GR-HorizonUniversalization

BH / Rindler / Cosmological / Acoustic Horizons as One Substrate Object

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Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of the four-horizon structural unification (BH / Rindler / cosmological / acoustic) from nothing. Result is conditional on the 13 ED primitives (Paper_087) + declared paper-specific postulates (SCBU, V5-saturation mechanism).
2. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage ('t Hooft cellular-automaton, Wolfram Ruliad, causal-set program), not the operational-reconstruction lineage.
3. It does **not** override Paper_095's three-tier verdict grammar. Verdict for this paper is stated in §1.
4. It does **not** claim derivation of the Hawking-Unruh temperature formula $T = \kappa/(2\pi)$ — that result is INHERITED from standard semi-classical GR (Bekenstein 1973, Hawking 1975, Gibbons-Hawking 1977, Unruh 1976). ED's substrate contribution is the V5-saturation mechanism identified as common microstructure across the four horizon types, NOT the temperature formula itself.
5. **INHERITED vs FORCED breakdown:** V5-saturation as common microstructure mechanism is the FORM-FORCED ED contribution; the universal Hawking-Unruh temperature formula and specific surface gravities / temperatures per horizon type are INHERITED standard physics.

Abstract

This paper supplies the substrate-level structural identification: the four standardly-distinguished horizon types — (i) black-hole event horizons, (ii) Rindler acceleration horizons, (iii) cosmological de-Sitter horizons, (iv) acoustic/dumb horizons in fluid analogues — are projections of the same ED substrate object: the V5-saturated boundary at which cross-chain bandwidth Γ_{cross} collapses (Paper_039). The four types differ in which parameter axis (mass / acceleration / Hubble rate / flow gradient) parameterizes the projection. The universal Hawking-Unruh temperature $T = \hbar\kappa/(2\pi k_B c)$ at each horizon is **INHERITED** from standard semi-classical GR; ED's contribution is the **common V5-saturation microstructure**. Per Paper_095 form-FORCED / value-INHERITED methodology, verdict is **M3** for the V5-saturation mechanism identification. Key falsifier **F2**: substrate evidence that V5 cross-chain saturation does NOT occur at one of the four horizon types.

1 Statement of Result

Claim. Four standardly-distinguished horizon types — (i) black-hole event horizons, (ii) Rindler acceleration horizons, (iii) cosmological de-Sitter horizons, (iv) acoustic/dumb horizons in fluid

analogues — are **projections of the same ED substrate object**: the V5-saturated substrate boundary at which cross-chain bandwidth Γ_{cross} collapses (Paper_039) and mobility on substrate cells degenerates.

The four types differ in **which parameter axis** parameterizes the projection:

- BH horizon: local gravitational mass $M \rightarrow r_H = 2GM/c^2$.
- Rindler horizon: local proper acceleration $\alpha \rightarrow \xi_R = c^2/\alpha$.
- Cosmological horizon: global Hubble rate $H_0 \rightarrow R_H = c/H_0$.
- Acoustic horizon: local flow velocity reaching local sound speed c_s .

Under SCBU + Paper_ED_SC_4.1's projection framework, all four sit in Regime 4 (cosmological-regime / boundary). The structural equivalence is form-IDENTIFIED. Specific numerical values per horizon type are INHERITED.

Verdict: **M3** — the structural identification is forced by V5-saturation + SCBU; the four horizon types are unified at substrate level, not merely analogically related.

2 Primitive Inputs

2.1 Primitives invoked

- **P10 (Rule-type primitive)** — V5 cross-chain kernel.
- **P11 (Commitment-irreversibility)** — directional substrate flow producing one-sided horizon structure.
- **P13 (Time homogeneity)** — temporal structure on which horizon-crossing rates are defined.

2.2 Upstream Dependencies

- **I-017**: Substrate→continuum Lorentz covariantization $\rightarrow$ acoustic metric.
- **I-039**: Horizon as decoupling surface $\Gamma_{\text{cross}} \rightarrow 0$ (Paper_039).
- **I-040**: Trans-Planckian resolution via V5 cutoff (Paper_040).
- **I-047**: Hawking spectrum at $T_H = \kappa/(2\pi)$ via substrate-Unruh / KMS.
- **I-028**: Cosmic decoupling surface $R_H = c/H_0$.
- **I-090**: V5 finite-bandwidth kernel.
- **I-SCBU**: Substrate-Cosmology Boundary Unification.
- **I-ED_SC_4_1**: BH $\leftrightarrow$ cosmic boundary projection.
- **I-Unruh**: Standard Unruh-effect machinery.

3 Derivation

3.1 The four standardly-distinguished horizon types

Black-hole event horizon. For a Schwarzschild black hole of mass M , the event horizon is the null hypersurface at $r = 2GM/c^2$. Observers crossing it cannot communicate back to spatial infinity. Surface gravity $\kappa = c^4/(4GM)$.

Rindler horizon. For a uniformly-accelerated observer in Minkowski spacetime, the Rindler horizon is the null hypersurface at coordinate distance $\xi_R = c^2/\alpha$ behind the observer. Causally inaccessible region for the accelerated observer. Surface gravity $\kappa_R = \alpha$.

Cosmological de-Sitter horizon. In de Sitter spacetime, the cosmological horizon is at $R_H = c/H_0$ for the comoving observer. Spatial regions beyond R_H recede causally; communication ceases. Surface gravity $\kappa_{dS} = cH_0$.

Acoustic/dumb horizon. In a fluid where flow velocity reaches the local sound speed c_s , the surface where $v = c_s$ acts as an acoustic horizon: sound waves cannot propagate upstream past this surface. Surface gravity $\kappa_{ac} = (1/2)|dv^2/dx|$ at the horizon.

3.2 Common substrate-level mechanism: V5 saturation

By Paper_039 + Paper_040 + Paper_ED_SC_4.1, the BH horizon is structurally identified as the substrate surface where V5 cross-chain bandwidth Γ_{cross} collapses to zero. Substrate participation cannot propagate across this surface; one-sided causal disconnection results.

This mechanism extends to all four horizon types:

Rindler horizon: From the perspective of the accelerated observer, V5 cross-chain content beyond the Rindler horizon is causally inaccessible. The substrate V5 budget for cross-horizon correlations vanishes (relative to the accelerated frame). Same V5-saturation structure, parameterized by acceleration α rather than mass M .

Cosmological horizon: V5 cross-chain content between regions separated by more than $R_H = c/H_0$ is causally disconnected (cosmological-redshift / cosmic expansion produces the decoupling). Same V5-saturation structure, parameterized by H_0 globally rather than M locally.

Acoustic horizon: In fluid analogues, the substrate-level mobility (V5-bandwidth content) vanishes across the surface where flow v reaches sound speed c_s . Phonons cannot propagate upstream; V5 cross-chain bandwidth for phonon-class excitations collapses.

In all four cases, the mechanism is **V5 cross-chain saturation**: substrate participation cannot maintain coherent cross-boundary correlations.

3.3 SCBU framework: same substrate object, four parameter axes

Under Paper_ED_SC_4.1 and Paper_ED_SC_4.6, the four horizon types are placed in Regime 4 (cosmological-boundary regime) of the unified four-regime RG structure. Each is a projection of the same substrate-cosmology boundary onto a different parameter axis:

Horizon	Parameter axis	Horizon scale	Surface gravity
BH	Local mass M	$r_H = 2GM/c^2$	$\kappa_{BH} = c^4/(4GM)$
Rindler	Local acceleration α	$\xi_R = c^2/\alpha$	$\kappa_R = \alpha$
Cosmological	Global rate H_0	$R_H = c/H_0$	$\kappa_{dS} = cH_0$
Acoustic	Local flow gradient	$v(x_H) = c_s$	$\kappa_{ac} = (1/2) dv^2/dx $

Substrate-level, all four are the **same boundary** — V5-saturated cross-chain disconnection — projected onto different parameter axes.

3.4 Universal substrate-Unruh / Hawking temperature

Attribution clarity: The universal Hawking-Unruh temperature formula $T = \kappa/(2\pi)$ for any Killing horizon (black-hole, cosmological, Rindler) is a result of standard semi-classical GR (Bekenstein 1973, Hawking 1975, Gibbons-Hawking 1977, Unruh 1976) and is INHERITED

here. ED’s genuine substrate contribution is the **V5-saturation mechanism** identified as the common microstructure across the four horizon types, not the temperature formula itself.

By Paper_047 (Hawking spectrum via substrate-Unruh / KMS), the temperature seen by an observer crossing or located near a horizon is universally

$$T = \frac{\hbar\kappa}{2\pi k_B c},$$

where κ is the surface gravity. The substrate-level derivation:

- Substrate KMS condition at the V5-saturated boundary.
- Imaginary-time periodicity = $2\pi/\kappa$.
- Bose-distribution spectrum at the corresponding temperature.

This same derivation applies to **all four horizon types**: BH gives Hawking radiation at T_H ; Rindler gives Unruh radiation at $T_U = \hbar\alpha/(2\pi k_B c)$; cosmological gives Gibbons-Hawking radiation at $T_{dS} = \hbar H_0/(2\pi k_B)$; acoustic gives “phonon Hawking radiation” at the analogous temperature for the fluid.

The universal substrate-Unruh / Hawking temperature is form-IDENTIFIED. The four phenomenological outcomes are projections of one substrate spectrum.

3.5 Why four classes and not more

The four horizon types listed are the standard ones. Other classes — Killing horizons in stationary spacetimes, apparent horizons, isolated horizons, anti-de-Sitter conformal-boundary horizons — fit within the same V5-saturation framework but require additional structural content (e.g., AdS boundary conditions for the anti-de-Sitter case).

The structural claim: **any** substrate-level V5-saturated boundary (per the SCBU framework) constitutes a horizon. The four listed types are the canonical instances; others are admissible extensions.

3.6 Geometric equivalence under ED rule-types

At substrate level, the four horizon types have geometric equivalence in the following structural sense:

- Each is a codimension-1 surface in the substrate-coarse-grained acoustic-metric manifold.
- Each is null with respect to the appropriate acoustic-metric / effective-metric.
- Each carries surface gravity κ inherited from V5 saturation rate.
- Each produces universal thermal radiation at $T = \hbar\kappa/(2\pi k_B c)$.
- Each obeys a substrate-level “first law” relating energy / area / temperature.

This is the **structural geometric equivalence** — not a metric-isometry equivalence, but a structural-feature equivalence under ED rule-types (V5 + DCGT + acoustic metric).

3.7 What is FORM-FORCED vs VALUE-INHERITED

form-IDENTIFIED:

- V5-saturation as universal horizon mechanism.
- Substrate-Unruh / Hawking universal-temperature formula.

- Four parameter axes (local mass / local acceleration / global rate / local flow gradient) as the projection coordinates.
- Codimension-1 null-surface structure.

VALUE-INHERITED:

- Specific numerical surface gravities (depend on M, α, H_0, v).
- Specific Hawking/Unruh/Gibbons-Hawking temperatures.
- Specific evaporation rates.

4 Audit Table

#	Step	Label	Notes
1	P10, P11, P13 primitives	P	Primitives.
2	BH horizon as $\Gamma_{\text{cross}} \rightarrow 0$ V5-sat.	I	Paper_039.
3	V5 cross-chain kernel	I	Paper_090.
4	Rindler V5-saturation in accelerated frame	D-via-I	Composition + Rindler.
5	Cosmological V5-saturation	I	Paper_028 + 4.1.
6	Acoustic V5-saturation for phonons	D-via-I	Acoustic analogue + V5.
7	SCBU places all four in Regime 4	I	Paper_ED_SC_4.1, 4.6.
8	Universal $T = \hbar\kappa/(2\pi k_B c)$	I	Paper_047.
9	Four-horizon structural identification	A $\rightarrow$ position	V5-saturation mechanism is the ED content.
10	Numerical surface gravities per type	I	Standard physics.
11	Specific temperatures	I	Standard physics.
12	Hawking-Unruh formula $T = \kappa/(2\pi)$	I	Bekenstein/Hawking/Gibbons/Unruh.
13	Verdict M3	A $\rightarrow$ position	Per Paper_095 §3.3.

5 Falsification Criteria

- **F1:** Empirical detection of horizon-class object NOT exhibiting universal $T = \hbar\kappa/(2\pi k_B c)$ thermal spectrum. Would refute the universalization.
- **F2:** Substrate evidence that V5 cross-chain saturation does NOT occur at one of the four horizon types. Would refute the universal mechanism.
- **F3:** Discovery of a fifth horizon class not derivable from V5-saturation. Would extend the framework (not refute).
- **F4:** SCBU postulate refuted — propagates from Paper_SCBU.

6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

This paper supplies the substrate-level universalization of horizon-class objects across BH, Rindler, cosmological, and acoustic types.

Relationship to other papers:

- **Paper__039 BH-horizon:** the canonical instance from which the framework extends.
- **Paper__028, 038:** cosmological-horizon companion.
- **Paper__047:** Hawking spectrum universal formula.
- **Paper__SCBU + 4.1 + 4.6:** SCBU framework placing all four in Regime 4.

Numerical content INHERITED. Surface gravities, temperatures per type. **Form IDENTIFIED.** Four-horizon structural identification + universal substrate-Unruh / Hawking temperature.

Verdict: **M3** — genuine ED contribution is the V5-saturation mechanism (common microstructure across the four horizon types); the universal temperature formula itself is INHERITED standard semi-classical GR.